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Overview of batteries State of Charge estimation methods

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Abstract

In the present, electric and hybrid electric vehicles are more and more common. Usually, the main source of energy are batteries. To achieve optimal utilization and protection of batteries, battery management is used. This battery management system prevents batteries from overdischarge and overcharge and provides cell balancing. Accurate state of charge estimation is necessary for these tasks. There are several methods for state of charge determination which can be divided as direct measurement methods, book-keeping methods, adaptive methods and hybrid methods. In this article, commonly used state of charge estimation methods, advantages and disadvantages are described.

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1. Introduction

Electric vehicles (EV) and hybrid electric vehicles (HEV) and plug-in hybrid electric vehicle (PHEV) are now more and more popular. The main element of all these vehicles is battery pack. To avoid of shortening battery life or even damaging of battery, batteries should not be over-discharged or overcharged. For battery protection and optimal utilization of battery is battery management system (BMS) is used. One of key component of battery management system is accurate state of charge (SOC) estimation because SOC reflect battery performance. Due to battery is storage of chemical energy and this chemical energy is not directly accessed, is SOC estimation very difficult.

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The SOC is one of the most important parameters for batteries, but its definition presents many different issues. In general, the SOC of a battery is defined as the ratio of its current capacity ($Q(t)$) to the nominal capacity (Q_n). The nominal capacity is given by the manufacturer and represents the maximum amount of charge that can be stored in the battery. The SOC can be defined as follows: [1]

$$SOC(t) = \frac{Q(t)}{Q_n} \quad (1)$$

2. SOC estimation methods

As mentioned in the introduction, the determination of battery SOC is always an essential part of a BMS. The accurate and reliable estimation of battery SOC can provide a necessary assessment factor for vehicle energy management and optimal design of the control system. Therefore, a larger number of methods have been proposed for estimating battery SoC in real-time. For comparing these methods in more detail, we have classified them into four group [2].

Table 1. Overview of SOC estimation methods

Category	Method
Direct measurement	Open circuit voltage method
	Terminal voltage method
	Impedance method
	Impedance spectroscopy method
Book-keeping estimation	Coulomb counting method
	Modified Coulomb counting method
Adaptive systems	Neural network
	Support vector machine
	Fuzzy neural network
	Kalman filter
Hybrid methods	Coulomb counting and EMF combination
	Coulomb counting and Kalman filter combination
	Per-unit system and EKF combination

3. Direct measurement methods

Direct measurement methods use some battery physical attributes such as impedance or terminal voltage. The most common methods are open circuit voltage method, terminal voltage method, impedance measurement method, impedance spectroscopy method.

3.1. Open circuit voltage method

There is approximately a linear relationship between the SOC of the lead-acid battery and its open circuit voltage (OCV) given by

$$VOC(t) = a1 \times SOC(t) + a0, \quad (2)$$

where $SOC(t)$ is the SOC of the battery at t , a_0 is the battery terminal voltage when $SOC = 0\%$, and a_1 is obtained from knowing the value of a_0 and $VOC(t)$ at $SOC = 100\%$. By (1), the estimation of the SOC is equivalent to the estimation of its OCV [8]. The OCV method based on the OCV of batteries is proportional to the SOC when they are disconnected from the loads for a period longer than two hours. However, such a long disconnection time may be too long to be implemented for battery [4]. Unlike the lead-acid battery, the Li-ion battery does not have a linear relationship between the OCV and SOC [5,6]. The OCV relationship with SOC was determined from applying a pulse load on the Li-ion battery, then allowing the battery to reach equilibrium. The relationship between the OCV and SOC cannot be exactly the same for all batteries. Because the conventional OCV-SOC differs among batteries, there is a problem in that the relationship of the OCV-SOC should be measured to estimate accurately the SOC [1].

3.2. Terminal voltage method

The terminal voltage method is based on the terminal voltage drops because of the internal impedances when the battery is discharging, so the electromotive force (EMF) of battery is proportional to the terminal voltage. Since the EMF of battery is approximately linear proportional to the SOC, the terminal voltage of battery is also approximately linear proportional to the SOC. The terminal voltage method has been employed at different discharge currents and temperatures [7]. But at the end of battery discharge, the estimated error of terminal voltage method is large, because the terminal voltage of battery suddenly drops at the end of discharge [1].

3.3. Impedance method

The method uses battery voltage and current to measure the internal resistance of the battery. Voltage is measured with the variation of current change during small duration (< 10 ms). The ratio of voltage and current variation results in DC resistance, which represents the capacity of the battery in DC. A small interval less than 10 ms is needed not only to capture the ohmic effect, but also to reduce the effect of transfer reaction and acid diffusion [8]. However, the value of estimated resistance contains an error if the time is longer. In addition, the method has good adaptability as well as the high accuracy of SOC estimation only during the end period of discharging. Furthermore, due to its low value (in milliohm range), the accuracy to get internal resistance is very hard to obtain. The internal resistance changes slightly with a wide range of SOC which is difficult to observe [9]. Due to this shortcoming, the DC internal resistance is hardly used to estimate SOC [10].

3.4. Impedance spectroscopy method

The impedance spectroscopy method measures battery impedances over a wide range of ac frequencies at different charge and discharge currents. The values of the model impedances are found by least-squares fitting to measured impedance values. SOC may be indirectly inferred by measuring present battery impedances and correlating them with known impedances at various SOC levels [1,11].

4. Book-Keeping methods

The Book-keeping estimation method uses battery discharging current data as input. This method permits to include some internal battery effects as self-discharge, capacity-loss, and discharging efficiency. Two kinds of book-keeping estimation methods have been employed: Coulomb counting method and modified Coulomb counting method [1].

4.1. Coulomb counting method

The Coulomb counting method measures the discharging current of a battery and integrates the discharging current over time in order to estimate SOC [12]. Coulomb counting method is done to estimate the $SOC(t)$, which is estimated from the discharging current, $I(t)$, and previously estimated SOC values, $SOC(t-1)$. SOC is calculated by the following equation:

$$\text{SOC}(t) = \text{SOC}(t-1) + I(t)n\Delta t. \quad (3)$$

But there are several factors that affect the accuracy of Coulomb counting method including temperature, battery history, discharge current, and cycle life [1, 12].

4.2. Modified Coulomb counting method

To improve the Coulomb counting method, a new technique called modified Coulomb counting method is proposed. The modified Coulomb counting method uses the corrected current to improve the accuracy of estimation. The corrected current is the function of discharging current. There is a quadratic relationship between the corrected current and discharging current of battery. By practice of experimental data, corrected current is calculated by the following form:

$$I_c(t) = k_2(t)^2 + k_1 I(t) + k_0, \quad (4)$$

where k_2 , k_1 and k_0 are constant values obtained from the practice experimental data. In modified Coulomb counting method, SOC is calculated by the following equation:

$$\text{SOC}(t) = \text{SOC}(t-1) + I_c(t)Qn\Delta t. \quad (5)$$

The experimental results show that the accuracy of the modified Coulomb counting method is superior to the conventional Coulomb counting method [1].

5. Adaptive systems

Recently, with the development of artificial intelligence, various new adaptive systems for SOC estimation have been developed. The new developed methods include back propagation (BP) neural network, radial basis function (RBF) neural network, fuzzy logic methods, support vector machine, fuzzy neural network, and Kalman filter. The adaptive systems are self-designing ones that can be automatically adjusted in changing systems. As batteries have been affected by many chemical factors and have nonlinear SOC, adaptive systems offer good solution for SOC estimation [1].

5.1. Neural network

A Neural Network uses a mathematical algorithm model for Complex Neural Network characteristics or parallel process. The Neural Network can achieve processing data and solve relations between various initial complex factors¹⁶. A BP Neural Network which is one of Neural Network algorithms solves a non-linear system and has a simple topology structure compared with the typical Neural Network methods [13]. The structure of the BP Neural Network is composed of three layers including input layer, hidden layer and output layer. The input layer includes battery voltage, current, resistance and ambient temperature. The number of the hidden layer depends on the system accuracy. The output layer of this system produces an estimated SOC value. The goal of this method is to reduce the minimum error value. However, the system error depends on the number of training data and experiment methods. The training data that are used to estimate SOC can be obtained from the experiments of charging and discharging batteries. This training method typically minimizes error functions. Thus, these errors increase if the BP Neural Network method uses not enough trained information from SOC values [14]. Therefore, training data from a large amount of batteries should be used for obtaining an accurate SOC value. This is because a battery discharge characteristics may differ from the amount of the electrolyte in batteries although their type and manufacturer are same [15].

5.2. Support vector machine

The support vector machine (SVM) has been applied for classification in various domains of pattern recognition. The SVM has also been applied for regression problem, even the regression problem inherently more difficult than classification problem. The SVM used as a nonlinear estimation system is more robust than a least squares estimation system because it is insensitive to small changes [1].

5.3. Fuzzy neural network

Fuzzy neural network (FNN) has been used in many applications, especially in identification of unknown systems. In nonlinear system identification, FNN can effectively fit the nonlinear system by calculating the optimized coefficients of the learning mechanism.

Lee et al. investigated a soft computing technique for estimating battery SOC of individual batteries in a battery string. The soft computing approach uses a fusion of an FNN with B-spline membership functions and a reduced-form genetic algorithm [1].

5.4. Kalman filter

Using real-time measurement road data to estimate the SOC of battery would normally be difficult or expensive to measure. Application of the Kalman filter method is provide verifiable estimations of SOC for the battery via the real-time state estimation. Yatsui and Bai [16] presented a Kalman filter based SOC estimation method for lithium-ion batteries. Experimental results validate the effectiveness of Kalman filter during the online application. Barbarisi et al. [17] presented an extended Kalman filter (EKF) to estimate the concentrations of the main chemical species which are averaged on the thickness of the active material in order to obtain the SOC of the battery, by using the terminal current and voltage measurements. Based on unscented Kalman filter (UKF) theory and a comprehensive battery model, a novel SOC estimation method is proposed in [18]. The results show that UKF method is superior to extended Kalman filter method in SOC estimation for battery. Sun et al. [19] presented an adaptive UKF method to estimate SOC of a lithium-ion battery for battery electric vehicles. The adaptive adjustment of the noise covariance in the SOC estimation process is implemented by an idea of covariance matching in the UKF context [1].

6. Hybrid methods

The object of hybrid models is to benefit from the advantages of each method and obtain a globally optimal estimating performance. Since the information contained in the individual estimating method is limited, hybrid method can maximize the available information, integrate individual model information, and make the best use of the advantages of multiple estimating methods thus improving the estimation accuracy. The literatures show that the hybrid methods generally produce good SOC estimating results compared to individual methods. The hybrid methods combine different approaches such as direct measurement method and book-keeping estimation method [1].

6.1. Coulomb counting and EMF combination

A new SOC estimation method that combines direct measurement method with the battery EMF measurement during the equilibrium state and book-keeping estimation with Coulomb counting method during the discharge state has been developed and implemented in a real-time estimation system [20]. Any battery will lose capacity during cycling. In order to calculate SOC and remaining run-time (RRT) accurately and to improve the SOC estimation system capability to cope with the aging effect, a simple Qmax adaptation algorithm is introduced. In this algorithm the stable conditions of the charge state are exploited in order to adapt Qmax with the aging effect. This paper has proved that the Qmax adaptation algorithm can improve the SOC and RRT estimation accuracy even for a fresh

battery. Since a battery loses capacity during cycling, it is concluded that the Qmax adaptation algorithm will increase substantially the SOC and the RRT estimation accuracy [1].

6.2. Coulomb counting and Kalman filter combination

New SOC estimation method was proposed denoted as “KalmanAh method,” which uses the Kalman filter method to correct for the initial value used in the Coulomb counting method. In Kalman Ah method, the Kalman filter method is used to make the approximate initial value converge to its real value. Then the Coulomb counting method is applied to estimate the SOC for the long working time. The SOC estimation error is 2.5% when compared with the real SOC obtained from a discharge test. This compares favorably with an estimation error of 11.4% when using Coulomb counting method [1].

6.3. Per-unit system and EKF combination

Kim and Cho described the application of an EKF combined with a per-unit (PU) system to the identification of suitable battery model parameters for the high accuracy SOC estimation of a lithium-ion degraded battery. To apply the battery model parameters varied by the aging effect, based on the PUsystem, the absolute values of the parameters in the equivalent circuit model in addition to the terminal voltage and current are converted into dimensionless values relative to a set of base value. The converted values are applied to dynamic and measurement models in the EKF algorithm [1].

4. Conclusion

In this article, common methods of estimation state of charge of batteries are described. Direct measurement methods are suitable for simply application because they require very low computation power, but accuracy in some intervals of SOC is very poor. OCV method is easy to implement, but for observe SOC, battery cannot be discharge so for traction batteries only when vehicle is not moving. These methods are most often used for lead acid batteries. Book-keeping methods which used current as input data, also need low computation power but they have cumulative error in integration of current. This method has very good results with lithium batteries. To eliminate this cumulative error, book-keeping methods are combined with other methods. Advantages of adaptive methods of estimation are high accuracy at full range of SOC, high accuracy at different discharge current but main disadvantage is high computation power for complex calculation. Also, linearization error could occur on non-linear systems. Accuracy of model-based method depending of complexity and accuracy of battery model.

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References

- [1] Wen-Yeau Chang, 2013. The state of Charge Estimating Methods for Battery: A review. Hindawi Publishing Corporation ISRN Applied Mathematics
- [2] RUI XIONG, JIAYI CAO, QUANQING YU, HONGWEN HE, AND FENGCHUN SUN, 2018. Critical Review on the Battery State of Charge Estimation Methods for Electric Vehicles. SPECIAL SECTION ON BATTERY ENERGY STORAGE AND MANAGEMENT SYSTEMS IEEE Access
- [3] New York, pp. 387.
- [3] J. Chiasson and B. Vairamohan, 2005. Estimating the state of charge of a battery, IEEE Transactions on Control Systems Technology, vol. 13, no. 3, pp. 465–470.
- [4] K. S. Ng, C. S. Moo, Y. P. Chen, and Y. C. Hsieh, December 2008. State-of-charge estimation for lead-acid batteries based on dynamic opencircuit

- voltage, Proceedings of the 2nd IEEE International Power and Energy Conference (PECon '08), pp. 972–976, Johor Bahru, Malaysia,
- [5] M. Coleman, C. K. Lee, C. Zhu, and W. G. Hurley, , 2007. State-of-charge determination from EMF voltage estimation: using impedance, terminal voltage, and current for lead-acid and lithium-ion batteries, *IEEE Transactions on Industrial Electronics*, vol. 54, no. 5, pp. 2550–2557.
- [6] T. Dong, J. Li, F. Zhao et al., May 2011. Analysis on the influence of measurement error on state of charge estimation of LiFePO₄ power battery,” in Proceedings of the International Conference on Materials for Renewable Energy and Environment (ICMREE '11), pp. 644–649, Shanghai, China.
- [7] A. H. Anbuky and P. E. Pascoe, 2000. VRLA battery state-of-charge estimation in telecommunication power systems,” *IEEE Transactions on Industrial Electronics*, vol. 47, no. 3, pp. 565–573
- [8] Wang HH, Liu YY., 2013. Estimation of state of charge of batteries for electric vehicles. *Int J Control Autom*;6:185–94.
- [9] Lu L, Han X, Li J, Hua J, Ouyang M., 2013. A review on the key issues for lithium-ion battery management in electric vehicles. *J Power Sources*;226:272–88.
- [10] M.A. Hannana, M.S.H. Lipub, A. Hussainb, A. Mohamedb, 2017, A review of lithium-ion battery state of charge estimation and management system in electric vehicle applications: Challenges and recommendations, *Renewable and Sustainable Energy Reviews* 78 (2017) 834–854
- [11] S. Rodrigues, N. Munichandraiah, and A. K. Shukla, 2000. A review of state-of-charge indication of batteries by means of A.C. Impedance measurements,” *Journal of Power Sources*, vol. 87, no. 1-2, pp. 12–20.
- [12] K. S. Ng, C. S. Moo, Y. P. Chen, and Y. C. Hsieh, 2009. Enhanced Coulomb counting method for estimating state-of-charge and state-of-health of lithium-ion batteries, *Applied Energy*, vol. 86, no. 9, pp. 1506–1511.
- [13] Dong C, Wang G., 2014. Estimation of power battery SOC based on improved BP neural network. 2014 IEEE International Conference on Mechatronics and Automation (ICMA); Tianjin. p. 2022–7.
- [14] Ehret Ch, Piller S, Schroer W, Jossen A., 2000. State-of-charge determination for lead-acid batteries in PV-applications. Proceedings of the 16th European Photovoltaic Solar Energy Conference; Glasgow;
- [15] Seonwoo Jeon¹, Jae-Jung Yun² and Sungwoo Bae¹, 2015. Comparative Study on the Battery State-of-Charge Estimation Method, *Indian Journal of Science and Technology*, Vol 8(26)
- [16] M. W. Yatsui and H. Bai, 2011. Kalman filter based state-of-charge estimation for lithium-ion batteries in hybrid electric vehicles using pulse charging, Proceedings of the 7th IEEE Vehicle Power and Propulsion Conference (VPPC '11), pp. 1–5, Chicago, Ill, USA.
- [17] O. Barbarisi, F. Vasca, and L. Glielmo, 2006. State of charge Kalman filter estimator for automotive batteries, *Control Engineering Practice*, vol. 14, no. 3, pp. 267–275.
- [18] J. Zhang and C. Xia, 2011. State-of-charge estimation of valve regulated lead acid battery based on multi-state unscented Kalman filter, *International Journal of Electrical Power & Energy Systems*, vol. 33, no. 3, pp. 472–476
- [19] F. Sun, X. Hu, Y. Zou, and S. Li, “Adaptive unscented Kalman filtering for state of charge estimation of a lithium-ion battery for electric vehicles,” *Energy*, vol. 36, no. 5, pp. 3531–3540, 2011.
- [20] V. Pop, H. J. Bergveld, P. H. L. Notten, J. H. G. Op het Veld, and P. P. L. Regtien, 2009. Accuracy analysis of the state-of-charge and remaining run-time determination for lithium-ion batteries, *Measurement*, vol. 42, no. 8, pp. 1131–1138.